

Tao2.4

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1 Exponential and Logarithm Functions

Define $\exp : \mathbb{R} \rightarrow \mathbb{R}$ by $\exp x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all $x \in \mathbb{R}$. Fix $x \in \mathbb{R} - 0$. $\lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{|x|^n} = \lim_{n \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1$, so $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges absolutely for all $x \in \mathbb{R}$ (since the series converges for $x = 0$), so $\exp$ is well-defined.

$\exp$ is analytic at 0 with $R = \infty$, so $\exp$ is analytic on $\mathbb{R}$. In particular, $\exp$ is differentiable and $(\exp x)' = \sum_{n=1}^{\infty} n \cdot \frac{x^{n-1}}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \exp x$.

Observe for $x \geq 0$, $\exp x \geq 1$. The sums $\sum_{n=0}^{\infty} \frac{x^n}{n!}, \sum_{m=0}^{\infty} \frac{y^m}{m!}$ are absolutely convergent for all $x, y \in \mathbb{R}$. Then write

$$\begin{aligned} \exp(x) \exp(y) &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \cdot \sum_{m=0}^{\infty} \frac{y^m}{m!} = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{1}{n!} \binom{n}{k} x^k y^{n-k} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (x+y)^n = \exp(x+y). \end{aligned}$$

So $\exp(x) \exp(-x) = \exp(0) = 1$, so $\exp$ is positive for all $x \in \mathbb{R}$. Take the restriction $\exp : \mathbb{R} \rightarrow \mathbb{R}^+$.

Note for all $x \geq 0$, we have $\exp(x) \geq 1$, so for $x \leq 0$, we have $\exp(x) \leq 1$. Since $(\exp x)' = \exp x \geq 1 = (x+1)'$ and $\exp(0) = 0+1$, we have $\exp x \geq x+1$. Similarly, $\exp x \geq x+1$ for $x \leq 0$, so $\exp x \geq x+1$ for all $x \in \mathbb{R}$.

Suppose $y \geq 1$. Then $\exp 0 \leq y = (y-1) + 1 \leq \exp(y-1)$. Since $\exp$ is analytic, $\exp$ is continuous, so by IVT, there exists $x \in [0, y-1]$ such that $y = \exp x$. If $0 < y < 1$, then $1/y = \exp(x)$, so $y = \exp(-x)$, so $\exp$ is surjective.

Note since $\exp x > 0$ for all $x \in \mathbb{R}$, $\exp$ is strictly increasing. Define $e = \exp 1$. I claim that $\exp x = e^x$ for all $x \in \mathbb{R}$.

Proof: Note if $f, g : X \rightarrow Y$ are continuous, $D \subset X$ is dense, Y is Hausdorff, and $f(x) = g(x)$ for all $x \in D$, then $f(x) = g(x)$ for all $x \in X$. In

particular, $\exp x, e^x$ are continuous. I claim that $\exp q = e^q$ for all rational q .

Let $q = m/n \geq 0$. Observe that $\exp(m/n)^n = \exp(m) = \exp(1)^m = e^m$ by an easy induction argument. Further, $\exp(m/n), e^m > 0$, so by definition, $\exp(m/n) = (e^m)^{1/n} = e^{m/n}$. It follows

$$\exp(-m/n) = \frac{1}{\exp(m/n)} = \frac{1}{e^{m/n}} = e^{-m/n},$$

so $\exp x, e^x$ agree for $x \in \mathbb{Q}$.

Define $\log : \mathbb{R}^+ \rightarrow \mathbb{R}$ to be the inverse of $\exp$, which exists and is continuous, increasing, bijective since $\exp$ has the same properties.

Inverse Function theorem. Let $f : (X, x_0) \rightarrow (Y, y_0)$ be bijective and x_0 a limit point of X . Let $g : Y \rightarrow X$ be the inverse of f such that g is continuous at y_0 . Suppose f is differentiable at x_0 and $f'(x_0) \neq 0$. Then y_0 is a limit point of Y , g is differentiable at y_0 , and $g'(y_0) = \frac{1}{f'(x_0)}$.

(1) y_0 is a limit point of Y . Let $\epsilon > 0$. Since f is diff at x_0 , f is cont at x_0 . Pick $\delta > 0$ such that $f(V_\delta(x_0) \cap X) \subset V_\epsilon(y_0)$. Since x_0 is a limit point of X , $V_\delta(x_0) \cap (X - x_0) \neq \emptyset$. Then

$$V_\epsilon(y_0) \cap (Y - y_0) \supset f(V_\delta(x_0) \cap (X - x_0)) \neq \emptyset,$$

so y_0 is indeed a limit point of Y .

I claim that

$$\lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{y \rightarrow y_0; y \in Y - y_0} \frac{y - y_0}{g(y) - g(y_0)}.$$

Let $\epsilon > 0$. Since f is differentiable at x_0 , there exists $\delta > 0$ such that $|f(x) - (f(x_0) + (x - x_0)f'(x_0))| \leq \epsilon|x - x_0|$ for all $x \in V_\delta(x_0)$. Since g is continuous at y_0 , pick $\delta' > 0$ such that $y \in V_{\delta'}(y_0) \rightarrow g(y) \in V_\delta(x_0)$. Then $|y - (y_0 + (g(y) - g(y_0))f'(x_0))| \leq \epsilon|g(y) - g(y_0)|$. For $y \neq y_0$, we have

$$\left| \frac{y - y_0}{g(y) - g(y_0)} - f'(x_0) \right| \leq \epsilon,$$

so $\lim_{y \rightarrow y_0; y \in Y - y_0} \frac{y - y_0}{g(y) - g(y_0)} = f'(x_0)$. Since $f'(x_0) \neq 0$, we can invert the limit:

$$\lim_{y \rightarrow y_0; y \in Y - y_0} \frac{g(y) - g(y_0)}{y - y_0} = g'(y_0) = \frac{1}{f'(x_0)}.$$

Since $(\exp x)' = \exp x > 0$ for all $x \in \mathbb{R}$, $\log$ is differentiable, and $(\log y)' = \frac{1}{(\exp x)'} = \frac{1}{y}$ for all $y \in \mathbb{R}^+$. $1/y$ is continuous, so by the Second Fundamental theorem,

$$\log x - \log 1 = \int_1^x \frac{dt}{t},$$

so $\log x = \int_1^x dt/t$.

Suppose $a, b > 0$. Since $\exp$ is surjective, let $a = \exp x, b = \exp y$ for $x, y \in \mathbb{R}$. Then

$$\log ab = \log \exp(x) \exp(y) = \log \exp(x+y) = x+y = \log a + \log b.$$

$$\log a^b = \log(e^x)^b = \log e^{xb} = xb = b \log a.$$

$$\log(1/a) = \log(1/\exp(x)) = \log \exp(-x) = -x = -\log a.$$

Consider the power series $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, which has $R = 1$ at $x = 0$. Then $-\log(1-x) = \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1} = \sum_{n=1}^{\infty} \frac{x^n}{n}$ has $R = 1$ at $x = 0$.

Make the substitution $x \mapsto 1-x$, so $-\log x = \sum_{n=0}^{\infty} \frac{(1-x)^n}{n}$, so $\log x = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n$ for $x \in (0, 2)$.

Claim: $\sum_{k=n+1}^{\infty} \frac{1}{k!} < \frac{1}{n!}$ for all $n \geq 1$. I claim that $(n+k)! \geq 2^k n!$ for all $k \geq 0, n \geq 1$. The result holds for $k = 0$. Suppose the result for k . Then $(n+k+1)! = (n+k+1)(n+k)! \geq 2 \cdot 2^k n! = 2^{k+1} n!$. It follows

$$\sum_{k=n+1}^{\infty} \frac{1}{k!} < \sum_{k=n+1}^{\infty} \frac{1}{n! 2^{k-n}} = \frac{1}{n!} \sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1}{n!}.$$

Let $a_n = n! \sum_{k=0}^n \frac{1}{k!} \in \mathbb{Z}^+$. We have for all $n \in \mathbb{Z}^+$ that $\frac{a_n}{n!} < e < \frac{a_{n+1}}{n!}$. Sps FTSOC $e = p/q$ for $p, q \in \mathbb{Z}^+$. Take $n = q$. Then $\frac{a_q}{q!} < \frac{p}{q} < \frac{a_{q+1}}{q!} \rightarrow a_q < p(q-1)! < a_q + 1$, absurd since $p(q-1)! \in \mathbb{Z}$.

Define

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

Then f is infinitely differentiable but not real analytic at 0.

Claim: Let P be a polynomial. Then $\lim_{x \rightarrow 0; x \in \mathbb{R}^+} P(1/x) e^{-1/x} = 0$. It suffices to show $\lim_{x \rightarrow 0; x \in \mathbb{R}^+} x^{-n} e^{-1/x} = 0$ for all $n \in \mathbb{Z}_{\geq 0}$ since polynomials are finite sums and limits are linear.

$e^{1/x} \geq (1/x)^{n+1}/(n+1)!$, so

$$x^{-n} e^{-1/x} \leq x^{-n} \cdot x^{n+1} \cdot (n+1)! = x(n+1)!.$$

By Squeeze lemma, $\lim_{x \rightarrow 0; x \in \mathbb{R}^+} x^{-n} e^{-1/x} = 0$.

Observe that for $x > 0$, $(P(1/x) e^{-1/x})' = P'(1/x) \cdot \frac{-1}{x^2} \cdot e^{-1/x} + P(1/x) \cdot e^{-1/x} \cdot$

$\frac{1}{x^2} = Q(1/x)e^{-1/x}$, where $Q(x) = x^2(P(x) - P'(x))$. Further, $\lim_{x \rightarrow 0; x \in \mathbb{R}^+} \frac{P(1/x)e^{-1/x}}{x} = \lim_{x \rightarrow 0; x \in \mathbb{R}^+} R(1/x)e^{-1/x} = 0$, where $R(x) = xP(x)$, so if g has the form

$$f(x) = \begin{cases} P(1/x)e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$$

then so does g' . Since f has this form, f is infinitely differentiable.

Observe that $f^{(n)}(0) = 0$ for all $n \in \mathbb{Z}_{\geq 0}$, so if f is real analytic at 0 with radius $r > 0$, then $f \equiv 0$ on $(-r, r)$, absurd since $e^{-1/(2r)} > 0$.

Sps $f : \mathbb{R} \rightarrow \mathbb{R}^+$ is diff such that $f'(x) = f(x)$ for all $x \in \mathbb{R}$. Then $f(x) = Ce^x$ for some $C > 0$.

Proof: Consider the function $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = \log f(x)$, well-defined since $f(\mathbb{R}) \subset \mathbb{R}^+$, and diff since $f, \log$ are diff on $\mathbb{R}, \mathbb{R}^+$, resp. By Chain rule, $g'(x) = f'(x)/f(x) = 1$, so $g(x) = x + c$ for some $c \in \mathbb{R}$. It follows $f(x) = \exp(x + c) = \exp(c)\exp(x) = Ce^x$ for some $C > 0$.

Consider the Geometric series $\sum_{n=0}^{\infty} (x/a)^n = \frac{1}{1-x/a} = \frac{a}{a-x}$ for $a > 0$ and $|x| < a$. Then

$$-a(\log(a-x) - \log a) = \sum_{n=1}^{\infty} \frac{1}{na^{n-1}} x^n,$$

so $\log(a-x) - \log a = -\sum_{n=1}^{\infty} \frac{1}{na^n} x^n$. Make the substitution $x \mapsto a-x$, so $\log x = \log a + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{na^n} (x-a)^n$ for $x \in (0, 2a)$ with radius of convergence a .

$e^{1/n} - 1 > 0$, so pick $N \in \mathbb{Z}^+$ such that $1/N < e^{1/n} - 1$. Then $(1 + 1/N)^n < e$, so $\alpha = e/(1 + 1/N)^n > 1$. For $x \geq N$, we have

$$\frac{e^{x+1}}{(x+1)^n} \cdot \frac{x^n}{e^x} = e/(1 + 1/x)^n \geq e/(1 + 1/N)^n = \alpha,$$

Let $M > 0$ denote the minimum of the function e^x/x^n on $[N, N+1]$ ($M > 0$ by EVT). For all $x \geq N$, write $x = m + a$ for $m \in \mathbb{Z}_{\geq N}$ and $a \in [0, 1]$. Then

$$\begin{aligned} \frac{e^x}{x^n} &= \left(\prod_{k=N}^{m-1} \frac{e^{k+1+a}}{(k+1+a)^n} \cdot \frac{(k+a)^n}{e^{k+a}} \right) \cdot \frac{e^{N+a}}{(N+a)^n} \\ &\geq \prod_{k=N}^{m-1} \alpha \cdot M = \alpha^{m-N} \cdot M. \end{aligned}$$

Since $\alpha > 1$, take $m \rightarrow \infty$, so $\frac{e^x}{x^n} \rightarrow \infty$.

The previous result gives $\lim_{x \rightarrow +\infty} \frac{P(x)}{e^{cx}} = \lim_{x \rightarrow +\infty} \frac{P(x/c)}{e^x} = 0$. Take $\epsilon = 1$, so

there exists $N \in \mathbb{Z}^+$ such that $x \geq N \rightarrow \frac{|P(x)|}{e^{cx}} < 1 \rightarrow |P(x)| < e^{cx}$.

Let $f : \mathbb{R}^+ \times \mathbb{R} \rightarrow \mathbb{R}^+$ by $f(x, y) = x^y$. Then f is cont.

Consider $g : \mathbb{R}^+ \times \mathbb{R} \rightarrow \mathbb{R}$ by $g(x, y) = \log f(x, y) = y \log x$. g is the composition

$$(x, y) \mapsto (\log x, y) \mapsto y \log x,$$

so g is cont. It follows $f(x, y) = \exp g(x, y)$ is cont as the composition of two cont functions.

2 Complex Numbers

Let (z_n) be a sequence of complex numbers. If $\sum_{n=0}^{\infty} |z_n|$ converges, so does $\sum_{n=0}^{\infty} z_n$.

$\mathbb{C}$ is complete, so it suffices to show the sequence of partial sums $(\sum_{n=0}^N z_n)_{n \in \mathbb{N}}$ is Cauchy. Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ such that $n, m \geq N \rightarrow \sum_{k=n}^m |z_k| < \epsilon$. Then

$$\left| \sum_{k=n}^m z_k \right| \leq \sum_{k=n}^m |z_k| < \epsilon.$$

Corr: If $(z_n), (w_n)$ are sequences such that $\sum_{n=0}^{\infty} |w_n|$ converges and $|z_n| \leq |w_n|$, then $\sum_{n=0}^{\infty} z_n$ converges.

Let z_n be a sequence such that $\alpha = \limsup_{n \rightarrow \infty} |z_n|^{1/n} < 1$. Then $\sum_{n=0}^{\infty} z_n$ converges.

Claim: Let (z_n) be a sequence of complex numbers. Let $\alpha = \limsup_{n \rightarrow \infty} |z_n|^{1/n}$. If $\alpha < 1$, then $\sum_{n=0}^{\infty} z_n$ is absolutely convergent. If $\alpha > 1$, then $\sum_{n=0}^{\infty} z_n$ is not convergent.

Let (a_n) be a sequence, $\alpha = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$. Suppose $\alpha < 1$. Let $\alpha' = (1 + \alpha)/2$, so $\alpha < \alpha' < 1$. Then $\alpha' > |a_n|^{1/n}$ for all $n \geq N$ for some $N \in \mathbb{N}$, so $|a_n| < (\alpha')^n$ for all $n \geq N$. The geometric series $\sum_{n=N}^{\infty} (\alpha')^n$ converges, so $\sum_{n=N}^{\infty} a_n$ and therefore $\sum_{n=0}^{\infty} a_n$ converges absolutely by Comparison test.

Suppose $\alpha > 1$. Then for all $N \in \mathbb{N}$, there exists $n \geq N$ such that $|a_n|^{1/n} > 1$, so we can construct a subsequence (a_{b_n}) of (a_n) such that $|a_{b_n}| > 1$ for all $n \in \mathbb{N}$, so (a_n) fails the zero test.

Suppose $\sum_{n=0}^{\infty} z_n$ converges. Then $\lim_{n \rightarrow \infty} z_n = 0$, which is equivalent to $\lim_{n \rightarrow \infty} |z_n| = 0$.

Ratio test: Let (z_n) be a sequence of nonzero complex numbers. If $\limsup_{n \rightarrow \infty} \frac{|z_{n+1}|}{|z_n|} < 1$, then $\sum_{n=0}^{\infty} z_n$ converges absolutely. If $\liminf_{n \rightarrow \infty} \frac{|z_{n+1}|}{|z_n|} > 1$, then $\sum_{n=0}^{\infty} z_n$ diverges.

Let $z \in \mathbb{C}$ be nonzero. Observe that

$$\limsup_{n \rightarrow \infty} \frac{|z|^{n+1}}{(n+1)!} \cdot \frac{n!}{|z|^n} = \limsup_{n \rightarrow \infty} \frac{|z|}{n+1} = 0,$$

so $\exp$ converges on $\mathbb{C}$.

I claim that $\exp(\bar{z}) = \overline{\exp z}$.

Proof: We have

$$\exp \bar{z} = \sum_{n=0}^{\infty} \frac{\bar{z}^n}{n!} = \lim_{N \rightarrow \infty} \sum_{n=0}^N \frac{\bar{z}^n}{n!} = \lim_{N \rightarrow \infty} \overline{\sum_{n=0}^N \frac{z^n}{n!}} = \overline{\lim_{N \rightarrow \infty} \sum_{n=0}^N \frac{z^n}{n!}} = \overline{\exp z}.$$

Define $\cos : \mathbb{C} \rightarrow \mathbb{C}$ by $\cos z = \frac{1}{2}(e^{iz} + e^{-iz})$, well-defined since $\exp$ converges on $\mathbb{C}$. Similarly, define $\sin : \mathbb{C} \rightarrow \mathbb{C}$ by $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$.

Note the equation $\exp(z+w) = \exp(z)\exp(w)$ for all $z, w \in \mathbb{C}$ (by rules of absolute convergence?). Then

$$\begin{aligned} \cos^2 z + \sin^2 z &= \left(\frac{e^{iz} + e^{-iz}}{2} \right)^2 + \left(\frac{e^{iz} - e^{-iz}}{2i} \right)^2 \\ &= \frac{1}{4} (e^{2iz} + 2 + e^{-2iz} - e^{2iz} + 2 - e^{-2iz}) \\ &= 1. \end{aligned}$$

If $x \in \mathbb{R}$, we can write

$$\begin{aligned} \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\ \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}. \end{aligned}$$

Observe that (for $x \neq 0$)

$$\lim_{n \rightarrow \infty} \frac{|(-1)^{n+1} x^{2n+2}|}{(2n+2)!} \cdot \frac{(2n)!}{|(-1)^n x^{2n}|} = \lim_{n \rightarrow \infty} \frac{|x|^2}{(2n+2)(2n+1)} = 0,$$

so $\cos$ has $R = \infty$ at $x = 0$. It follows $\cos$ is analytic on $\mathbb{R}$. Similarly,

$$\lim_{n \rightarrow \infty} \frac{|(-1)^{n+1} x^{2n+3}|}{(2n+3)!} \cdot \frac{(2n+1)!}{|(-1)^n x^{2n+1}|} = \lim_{n \rightarrow \infty} \frac{|x|^2}{(2n+3)(2n+2)},$$

so $\sin$ is analytic on $\mathbb{R}$.

We conclude that $(\cos x)' = -\sin x$ and $(\sin x)' = \cos x$.

Sum identities:

$$\begin{aligned}\cos z \cos w - \sin z \sin w &= \frac{1}{4}(e^{iz} + e^{-iz})(e^{iw} + e^{-iw}) + \frac{1}{4}(e^{iz} - e^{-iz})(e^{iw} - e^{-iw}) \\ &= \frac{1}{4}(e^{i(z+w)} + e^{i(-z+w)} + e^{i(z-w)} + e^{-i(z+w)} + e^{i(z+w)} - e^{i(-z+w)} - e^{i(z-w)} + e^{-i(z+w)}) \\ &= \frac{1}{2}(e^{i(z+w)} + e^{-i(z+w)}) = \cos(z+w).\end{aligned}$$

$$\begin{aligned}\cos z \sin w + \sin z \cos w &= \frac{1}{4i}(e^{iz} + e^{-iz})(e^{iw} - e^{-iw}) + \frac{1}{4i}(e^{iz} - e^{-iz})(e^{iw} + e^{-iw}) \\ &= \frac{1}{4i}(e^{i(z+w)} + e^{i(-z+w)} - e^{i(z-w)} - e^{-i(z+w)} + e^{i(z+w)} - e^{i(-z+w)} + e^{i(z-w)} - e^{-i(z+w)}) \\ &= \frac{1}{2i}(e^{i(z+w)} - e^{-i(z+w)}) = \sin(z+w).\end{aligned}$$

For $x \neq 0$,

$$\frac{\sin x}{x} = \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!},$$

so $\lim_{x \rightarrow 0; x \in \mathbb{R}} \frac{\sin x}{x} = 1$.

3 Rearrangement of Series

Let (a_n) be nonnegative such that $\sum_{n=0}^{\infty} a_n = L$ converges. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Then $\sum_{n=0}^{\infty} a_{f(n)} = L$.

Proof: Let $m \in \mathbb{N}$, so $\emptyset \subsetneq f([m]) \subset \mathbb{N}$. Let $M \in f([m])$ be maximal. Then

$$\sum_{n=0}^m a_{f(n)} = \sum_{n \in [m]} a_{f(n)} = \sum_{y \in f([m])} a_y \leq \sum_{y \in [M]} a_y = \sum_{n=0}^M a_n \leq L.$$

Hence, $\sum_{n=0}^{\infty} a_{f(n)}$ converges to $L' \leq L$. Conversely, since $a_n = a_{f^{-1}f(n)}$, we have $L \leq L'$, so $L = L'$.

Let (a_n) be a sequence such that $\sum_{n=0}^{\infty} |a_n|$ converges, so $\sum_{n=0}^{\infty} a_n = L \in \mathbb{R}$. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Then $\sum_{n=0}^{\infty} a_{f(n)} = L$.

Proof: Consider the sequence (b_n) given by $b_n = a_n + |a_n|$. Then $b_n \geq 0$ and $\sum_{n=0}^N b_n \leq 2 \sum_{n=0}^N |a_n| \leq 2L$, so $\sum_{n=0}^{\infty} b_n$ is absolutely convergent, so

$$\sum_{n=0}^{\infty} a_{f(n)} + \sum_{n=0}^{\infty} |a_{f(n)}| = \sum_{n=0}^{\infty} b_{f(n)} = \sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} |a_n|.$$

Since $\sum_{n=0}^{\infty} |a_{f(n)}| = \sum_{n=0}^{\infty} |a_n|$, we have $\sum_{n=0}^{\infty} a_{f(n)} = \sum_{n=0}^{\infty} a_n = L$ (how to generalize to $\mathbb{C}$?)

Let $S = \sum_{n=0}^{\infty} z_n$ be a series of complex numbers. Let $z_n = a_n + b_n i$. Then S is absolutely convergent iff $\sum_{n=0}^{\infty} a_n, \sum_{n=0}^{\infty} b_n$ are absolutely convergent.

(1). Suppose S is absolutely convergent, so $\sum_{n=0}^{\infty} |z_n| < \infty$. Since $|a_n| \leq |z_n|$, by Comparison test, $\sum_{n=0}^{\infty} |a_n| < \infty$. Similarly, $\sum_{n=0}^{\infty} |b_n| < \infty$.

(2). Suppose $\sum_{n=0}^{\infty} a_n, \sum_{n=0}^{\infty} b_n$ are absolutely convergent. Then

$$\sum_{n=0}^{\infty} \sqrt{a_n^2 + b_n^2} \leq \sum_{n=0}^{\infty} |a_n| + |b_n| < \infty + \infty = \infty,$$

so $\sum_{n=0}^{\infty} z_n$ is absolutely convergent.

Suppose $\sum_{n=0}^{\infty} z_n = L$ is absolutely convergent and $f : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection. Then $\sum_{n=0}^{\infty} z_{f(n)} = L$. Let $L = A + Bi$, so $\sum_{n=0}^{\infty} a_n = A$ and $\sum_{n=0}^{\infty} b_n = B$. Since $\sum_{n=0}^{\infty} z_n$ is absolutely convergent, so are $\sum_{n=0}^{\infty} a_n, \sum_{n=0}^{\infty} b_n$, so

$$\sum_{n=0}^{\infty} z_{f(n)} = \sum_{n=0}^{\infty} a_{f(n)} + i \sum_{n=0}^{\infty} b_{f(n)} = \sum_{n=0}^{\infty} a_n + i \sum_{n=0}^{\infty} b_n = A + Bi = L.$$

This justifies $\exp(z+w) = \exp(z)\exp(w)$ for $z, w \in \mathbb{C}$.

Let $E(x) = x - \sin x$. Then $E(0) = E'(0) = 0$. $E''(x) = \sin x$, so $|E''(x)| \leq 1$, so $|E'(x)| = |\int_0^x E''(t) dt| \leq \int_0^x |E''(t)| dt \leq \int_0^x 1 dt = x$. Then $|E(x)| = |\int_0^x E'(t) dt| \leq \int_0^x |E'(t)| dt \leq \int_0^x t dt = \frac{1}{2} t^2$.

In general, let $E(x) = \sum_{k=0}^n \frac{(-1)^k x^{2k+1}}{(2k+1)!} - \sin x$, so $E^{(0)}(0) = E^{(1)}(0) = \dots = E^{(2n+1)}(0)$ and $E^{(2n+2)}(x) = (-1)^n \sin x$, so $|E^{(2n+2)}(x)| \leq 1$. Integrating $2n+2$ times, by similar reasoning we have $|E(x)| \leq \frac{x^{2n+2}}{(2n+2)!}$.

Take $n = 4$, so we find $\sin x < 0$ for some $x \in \mathbb{R}^+$. Since $\sin 1 > 0$, by IVT there exists $x \in \mathbb{R}^+$ such that $\sin x = 0$. Denote by $Z = \sin^{-1}(0) \cap \mathbb{R}^+$. Note that $0 \notin \overline{Z}$, since otherwise there exists a sequence of points (x_n) such that $x_n \rightarrow 0$ and $\sin x_n = 0$ for all $n \in \mathbb{N}$, which forces $\sin \equiv 0$, absurd. Then $Z \subset [r, +\infty)$ for some $r > 0$, so $Z = [r, +\infty) \cap \sin^{-1}(0)$, so Z is closed.

Since $Z \neq \emptyset$ is bounded below by $r > 0$, let $\alpha = \inf Z \in Z$ be positive. Consider $\sin$ restricted to $[0, \alpha]$. Sps FTSOC $\sin x < 0$ for some $x \in (0, \alpha)$. Note that $\sin x \geq x - \frac{x^3}{6} - \frac{x^4}{24} \rightarrow \sin x > 0$ for $x \in [0, 2)$, so $\alpha \geq 2$. Note $\sin 1 > 0$, so $\sin$ is positive on $(0, \alpha)$.

We have the bounds $2 \leq \alpha \leq 4$.

$$\sin \alpha = 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} = 0,$$

so $\cos \frac{\alpha}{2} = 0$. Since $\sin^2 \frac{\alpha}{2} + \cos^2 \frac{\alpha}{2} = 1$ and $\sin \frac{\alpha}{2} > 0$, we have $\sin \frac{\alpha}{2} = 1$. It follows $\cos \alpha = 1 - 2 \sin^2 \frac{\alpha}{2} = -1$. Now

$$\sin(\alpha - x) = \sin \alpha \cos x - \cos \alpha \sin x = \sin x,$$

$$\sin(\alpha/2 - x) = \sin(\alpha/2) \cos x - \cos(\alpha/2) \sin x = \cos x,$$

$$\sin 2\alpha = 2 \cos \alpha \sin \alpha = 0,$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1,$$

$$\sin(2\alpha + x) = \sin 2\alpha \cos x + \cos 2\alpha \sin x = \sin x,$$

$$\sin(\alpha + x) = \sin \alpha \cos x + \cos \alpha \sin x = -\sin x.$$

Suppose $\sin x = 0$ for some $x \in \mathbb{R}$. Let $\frac{x}{\alpha} = n + a$ for $n \in \mathbb{N}$ and $a \in [0, 1)$. Then $x = \alpha(n + a)$, so by induction,

$$\sin x = \sin \alpha(n + a) = (-1)^n \sin(\alpha a) = 0.$$

Since $\alpha a \in [0, \alpha)$, we have $\alpha a = 0$, so $a = 0$. It follows $\sin x = 0$ iff $x = \alpha n$ for some $n \in \mathbb{N}$.

We can solve $\cos x = 0$ from $\cos x = \sin(\alpha/2 - x)$. The smallest positive real for which $\cos x = 0$ is $\alpha/2$. Since $\cos 0 = 1$, by IVT $\cos x > 0$ on $[0, \alpha/2]$, so $\sin$ is increasing on $[0, \alpha/2]$ and decreasing on $[\alpha/2, \alpha]$.

Consider the restriction of $\sin$ to $[-\alpha/2, \alpha/2]$, so $\sin$ is increasing, cont, so inverse $\sin^{-1} : [-1, 1] \rightarrow [-\alpha/2, \alpha/2]$ is cont increasing. If I take the restriction $\sin^{-1}$ to $(-\alpha/2, \alpha/2)$, $(\sin x)' = \cos x > 0$, so

$$(\sin^{-1})'(y) = \frac{1}{\cos x}.$$

We have $\cos^2 x + y^2 = 1$, so $\cos x = \sqrt{1 - y^2}$, so $(\sin^{-1})'(y) = \frac{1}{\sqrt{1 - y^2}}$.

Define $\tan : (-\alpha/2, \alpha/2) \rightarrow \mathbb{R}$ by $\tan(x) = \sin x / \cos x$, well-defined since $\cos x > 0$. Then $\tan'(x) = \frac{1}{\cos^2 x} > 0$, so $\tan$ is increasing cont.

Why is $\tan$ surjective? ISTS $\tan$ is unbounded above on $\mathbb{R}^+$. Write

$$\tan x = \frac{\sin x}{\sqrt{1 - \sin^2 x}}.$$

$\lim_{x \rightarrow 1; x \in (-1,1)} \frac{x}{\sqrt{1-x^2}} = \infty$. Since $\sin : (-\alpha/2, \alpha/2) \rightarrow (-1, 1)$ is bijective and inc, $\lim_{\theta \rightarrow \alpha/2; \theta \in (-\alpha/2, \alpha/2)} \tan \theta = \infty$.

Then we can define $\tan^{-1} : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$.

$$(\tan^{-1} y)' = \cos^2 x = \frac{\cos^2 x}{\sin^2 x + \cos^2 x} = \frac{1}{1 + \tan^2 x} = \frac{1}{1 + y^2}.$$

Sps FTSOC $\sin x \neq 0$ for all $x \in \mathbb{R}^+$. Then $\cos x \neq 0$ for all $x \in \mathbb{R}^+$. By IVT, $\sin, \cos > 0$ over $\mathbb{R}^+$. Consider $\cot : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. For $x \geq 1$, $\cot x = \cot 1 + \int_1^x \frac{-dt}{\sin^2 t} \leq \cot 1 + \int_1^x (-dt) = \cot 1 - x + 1$. Take $x \rightarrow \infty$, so $\cot x < 0$ for some $x \in \mathbb{R}^+$, absurd.

Consider the Power series

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n},$$

which converges absolutely if $|x| < 1$. Integrating both sides,

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}.$$

Note that the series converges at $x = 1$ by AST. Then by Abel's theorem, $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \lim_{x \rightarrow 1; x \in (-1,1)} \arctan x = \arctan 1 = \frac{\pi}{4}$, so

$$\pi = 4 \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}.$$

Let $f(x) = \sum_{n=1}^{\infty} 4^{-n} \cos(32^n \pi x)$.

Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f_n(x) = 4^{-n} \cos(32^n \pi x)$, so f_n is cont. Since $\sum_{n=0}^{\infty} 4^{-n} < \infty$ by Geometric series test, we have $f = \sum_{n=0}^{\infty} f_n$ converges uniformly, so f is cont.

We have

$$\begin{aligned} f\left(\frac{j+1}{32^m}\right) - f\left(\frac{j}{32^m}\right) &= \sum_{n=1}^{\infty} 4^{-n} \left[\cos\left(32^n \pi \frac{j+1}{32^m}\right) - \cos\left(32^n \pi \frac{j}{32^m}\right) \right] \\ &= 4^{-m} \Delta_m + 4^{-m} \sum_{k=1}^{\infty} 4^{-k} \cdot \Delta_{m+k}, \end{aligned}$$

where $\Delta_m \in \{-2, 2\}$ and $|\Delta_{m+k}| \leq 2$. We have

$$\left| \sum_{k=1}^{\infty} 4^{-k} \cdot \Delta_{m+k} \right| \leq \sum_{k=1}^{\infty} 4^{-k} \cdot 2 = \frac{1}{2} \cdot \frac{1}{1-1/4} = \frac{2}{3}.$$

It follows

$$\left| f\left(\frac{j+1}{32^m}\right) - f\left(\frac{j}{32^m}\right) \right| \geq 2 \cdot 4^{-m} - \frac{2}{3} \cdot 4^{-m} \geq 4^{-m}.$$

Sps FTSOC f is diff at $x_0 \in \mathbb{R}$. Pick $\delta > 0$ such that for $x \in V_\delta(x_0) - x_0$, we have $\frac{f(x)-f(x_0)}{x-x_0} \in V_1(f'(x_0))$, so $\frac{f(x)-f(x_0)}{x-x_0}$ is bounded for $x \in V_\delta(x_0) - x_0$. Pick $m \in \mathbb{Z}^+$ such that $32^{-m} < \delta$. Pick $k \in \mathbb{Z}$ such that $\frac{k}{32^m} \leq x_0 < \frac{k+1}{32^m}$. Then $\frac{k}{32^m} = \frac{k+1}{32^m} - \frac{1}{32^m} > x_0 - \delta$ and $\frac{k+1}{32^m} = \frac{k}{32^m} + \frac{1}{32^m} < x_0 + \delta$.

If $x = k/32^m$, we have

$$\frac{f((k+1)/32^m) - f(x_0)}{(k+1)/32^m - x_0} \geq 8^m \geq 8^{m-1}.$$

Else,

$$\frac{f((k+1)/32^m) - f(x_0)}{(k+1)/32^m - x_0} + \frac{f(x_0) - f(k/32^m)}{x_0 - k/32^m} \geq \frac{f((k+1)/32^m) - f(k/32^m)}{(k+1)/32^m - k/32^m} \geq 8^m,$$

so either way for each $m \in \mathbb{Z}^+$, there exists $x \in V_\delta(x_0) - x_0$ such that $\frac{f(x)-f(x_0)}{x-x_0} \geq 8^{m-1}$, absurd.